Categorical Encoding and Threshold Selection for BRFSS Classification

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***Abstract*—We classify self-reported myocardial infarction or coronary heart disease in BRFSS-derived survey data using NumPy implementations of linear models. We compare imputation, missingness indicators, categorical encoding and regularization on a stratified development/tuning split, reserving an independent final holdout. Reference encoding of small integer-valued feature sets improves tuning F1 by 0.0147 at a common threshold. Our selected L2 logistic model attains holdout F1 0.4225, accuracy 0.8685, and recall 0.5446 at a tuning-selected threshold of 0.20. Threshold choice dominates the measured accuracy–recall tradeoff. These observational, self-reported outcomes support classification comparisons, not prospective clinical claims.**

I. Data and evaluation design

The supplied dataset contains 328,135 labeled observations and 109,379 test observations, each with 321 features. There are 28,975 positive training labels (8.8302%). The target `_MICHD` describes respondents reporting a history of myocardial infarction or coronary heart disease, derived from the corresponding survey questions [1]. It does not establish prospective incidence or clinically verified diagnosis. Always predicting the negative class obtains 0.9117 accuracy but zero recall and F1.

Raw feature missingness is 44.7880% in training and 44.7852% in test. No infinite values, duplicate IDs, train/test ID overlap, or exact duplicate rows across all 321 raw features were found. Feature names and order match between splits, and label IDs align with training inputs. These checks do not establish that distinct rows represent independent respondents. Many survey codes have variable-specific meanings; for example, 88 denotes zero unhealthy days for `PHYSHLTH` [1]. We therefore do not apply a universal sentinel rule.

Using seed 433, we stratify labeled rows into 196,881 development, 65,627 tuning, and 65,627 untouched holdout observations. Every fill value, scale, category set, and feature-removal choice is fitted only on development rows during selection. Ten configurations and the selection rule were recorded before fitting. The AIcrowd rules specify accuracy, while the observed leaderboard lists F1 first and appears ordered by F1 [2], [3]. Pending clarification, we provisionally optimize F1 locally and report accuracy separately.

II. Methods

Feature processing. We remove the input ID and exclude annual sequence number `SEQNO`, its duplicate `_PSU`, and raw interview date `IDATE`. We also exclude `HAREHAB1`: its rehabilitation question is conditional on a reported heart attack [1]. All 773 observed training values occur in positive cases, making availability a target proxy. Other medical-history predictors are retained.

Numeric columns receive a development-fitted mean or median fill followed by mean/std scaling; fully observed constants and all-missing columns are removed. Optional binary indicators represent missingness observed in the fit partition. The categorical variant reference-encodes integer-valued columns with at most 12 observed levels, using the most frequent level as reference. Unseen categories map to that reference. This is a cardinality heuristic rather than a complete semantic interpretation; higher-cardinality survey codes remain numeric.

Learning. All six mandatory methods are implemented with the specified signatures: MSE GD, single-sample SGD, normal-equation least squares and ridge, and ordinary and regularized logistic GD. Their returned loss is evaluated at the final weights and excludes regularization. Ridge uses $X^\top X + 2N\lambda I$, penalizing every supplied coordinate. Independent checks supplement the official numerical tests.

For prediction we compare linear regression, ridge and logistic models. Our optional logistic optimizer minimizes

$$\frac{1}{N}\sum_i \left[\log(1+e^{x_i^\top w}) - y_i x_i^\top w\right] + \lambda \sum_{j>0} w_j^2, \qquad (1)$$

with $y_i \in \{0,1\}$ and an unpenalized intercept w_0. This intercept treatment is separate from the mandatory methods. A NumPy L-BFGS implementation uses ten curvature pairs and Armijo backtracking, at most 120 iterations, and a gradient infinity-norm target of 10^{-5}. Float32 design storage limits memory; weights, objective reductions and curvature state use float64. Several runs, including the selected model, reach the iteration budget rather than the target tolerance. The linear baseline handles redundant columns with a Gram eigen pseudoinverse; ridge uses a float64 normal-equation solve.

Selection. We compare the four processing variants in Table II, then $\lambda \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ on the categorical design. Thresholds range from 0.05 to 0.65 by 0.01, using tuning predictions only. Within 0.002 of the maximum tuning F1, the recorded rule prefers fewer features, then stronger L2 regularization. It selects $\lambda = 10^{-4}$ and threshold 0.20, with 1,172 design columns. Selection is frozen before holdout evaluation.

Table I
TUNING RESULTS. EACH MODEL'S THRESHOLD IS SELECTED ON TUNING DATA; THE MAJORITY CLASSIFIER IS FIXED. LINEAR MODELS USE THE INDICATOR DESIGN.

Model	Threshold	Accuracy	F1	Recall
Majority	—	0.9117	0.0000	0.0000
Linear regression	0.22	0.8653	0.4057	0.5208
Ridge (10^{-3})	0.22	0.8666	0.4049	0.5141
Logistic, mean fill	0.21	0.8724	0.4133	0.5089
Categorical, no L2	0.21	0.8763	0.4327	0.5343
Selected L2 logistic	0.20	0.8722	0.4321	0.5506

Table II
CONTROLLED TUNING ABLATIONS AT THRESHOLD 0.20 AND THE SAME 120-ITERATION BUDGET. L2 IS 10^{-3} UNTIL THE FINAL ROW. DELTAS ARE ABSOLUTE F1 UNITS.

Intervention	F1	ΔF1
Mean fill + scaling	0.4129	—
Median fill	0.4131	+0.0002
+ Missing indicators	0.4139	+0.0008
+ Categorical encoding	0.4285	+0.0147
L2 $10^{-3} \to 10^{-4}$	0.4321	+0.0036

III. MEASURED RESULTS

Table I compares tuning performance; these estimates are optimistic after selection. The unregularized categorical model achieves a slightly higher F1 (0.4327), within the prespecified tolerance for choosing regularization. Table II instead fixes the threshold at 0.20 to isolate processing changes. Categorical encoding provides the largest processing gain; median imputation and missing indicators have much smaller effects. Increasing L2 to 10^{-2} lowers tuned F1 to 0.4099, so stronger regularization is not uniformly beneficial. Equal optimizer budgets do not imply equal convergence, which limits attribution of small differences solely to feature representation.

On the untouched holdout, the fixed selection obtains F1 0.4225, accuracy 0.8685, balanced accuracy 0.7223, precision 0.3451, and recall 0.5446. There are 3,156 true positives, 5,989 false positives, 2,639 false negatives and 53,843 true negatives. ROC-AUC is 0.8585 and average precision is 0.3861. Training F1 is 0.4392 versus tuning 0.4321; the holdout gap remains a limitation rather than evidence of an unbiased tuning estimate.

At threshold 0.50, this same holdout model gives F1 0.2444 and accuracy 0.9142. Using the tuning-selected 0.20 threshold improves holdout F1 by 0.1781 while reducing accuracy by 0.0456; it increases recall from 0.1572 to 0.5446. Figure 1 shows the corresponding tuning tradeoff. An accuracy-selected threshold of 0.58 instead gives tuning accuracy 0.9153 and F1 0.1796, underscoring the unresolved competition objective.

Three-fold CV on the non-holdout pool yields F1 0.4310 $\pm$ 0.0004 (sample standard deviation). Each fold refits processing and calibrates its threshold inside the outer training partition. Family selection is not nested, so this is a consistency diagnostic. A conditional IID-row bootstrap interval for holdout F1 is [0.4125, 0.4321]; it ignores survey design and selection variability and is not a population-level clinical interval.

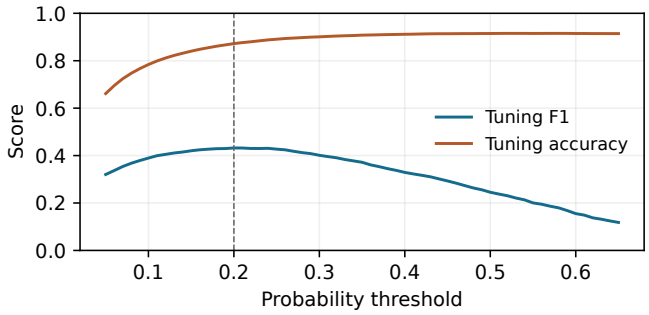


Figure 1. Tuning scores for the selected logistic model. The dashed line marks the tuning-selected threshold; no holdout labels determine this choice.

IV. LIMITATIONS AND REPRODUCTION

Survey nonresponse, optional modules, self-report and shared sampling structure limit interpretation. False negatives remain substantial (45.54% of positives), and a recall-oriented threshold introduces many false positives. We did not establish prospective validity, fairness across populations, full semantic cleaning, or coefficient-wise stability. Small ablation gains are not established as significant or consistent across folds; only the selected configuration has the three-fold diagnostic. The 120-step final fit also stops before convergence.

After freezing choices, we refit processing and weights on all labeled rows, producing 1,183 design columns because the larger fit discovers more categories. `run.py` loads the saved weights and fitted transformations; seed 433 and threshold 0.20 reproduce the CSV. From the repository root, run `python run.py` with `--data-dir dataset`, `--output new.csv`, and `--seed 433`. Two executions, including a fresh source directory and CSV cache, produce identical SHA-256 hashes in the same Python 3.12.14 / NumPy 2.5.3 environment. Cross-backend bitwise identity is unverified. Training code, indices, configurations, source/data fingerprints and measurements are saved. No AIcrowd acceptance or score is claimed.

Unresolved: public GitHub commit permalink, staff AI-use policy and metric clarification. The requester authorized AI assistance; independent course-policy verification remains pending. All students must review the generated code and text before submission.

REFERENCES

[1] CDC, "2015 BRFSS Codebook and Calculated Variables," 2016–2017. https://www.cdc.gov/brfss/annual_data/annual_2015.html.

[2] AIcrowd, "EPFL Machine Learning Project 1: Challenge Rules," accessed Oct. 29, 2026. https://www.aicrowd.com/challenges/epfl-machine-learning-project-1/challenge_rules.

[3] AIcrowd, "EPFL Machine Learning Project 1: Leaderboard," accessed Oct. 29, 2026. https://www.aicrowd.com/challenges/epfl-machine-learning-project-1/leaderboards.